VALVE COMPONENTS

42 component records | Numeric values are numbers without units in the cell. Units: diameter mm; pressure bar; flow m^3/h. All numeric specifications are sample values for software testing, not vessel-approved operating data.

FA546 — STOP VALVE (GLOBE)

id	FA546
component_name	FA546 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA546
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	50
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA546_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA546_IN, PIPE_FA546_OUT
relationships.related_component_ids	

FA582 — BALL VALVE

id	FA582
component_name	FA582 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA582
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA582_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA582_IN, PIPE_FA582_OUT
relationships.related_component_ids	

FA572 — BALL VALVE

id	FA572
component_name	FA572 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA572
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	80
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	24.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA572_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA572_IN, PIPE_FA572_OUT
relationships.related_component_ids	

FA512 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA512
component_name	FA512 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA512
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA512_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA512_IN, PIPE_FA512_OUT
relationships.related_component_ids	

FA542 — GATE VALVE

id	FA542
component_name	FA542 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA542
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	100
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	52.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA542_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA542_IN, PIPE_FA542_OUT
relationships.related_component_ids	

FA541 — GATE VALVE

id	FA541
component_name	FA541 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA541
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	125
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	20.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA541_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA541_IN, PIPE_FA541_OUT
relationships.related_component_ids	

FA511 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA511
component_name	FA511 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA511
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	150
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA511_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA511_IN, PIPE_FA511_OUT
relationships.related_component_ids	

FA581 — BALL VALVE

id	FA581
component_name	FA581 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA581
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	50
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	24.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA581_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA581_IN, PIPE_FA581_OUT
relationships.related_component_ids	

FA571 — BALL VALVE

id	FA571
component_name	FA571 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA571
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA571_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA571_IN, PIPE_FA571_OUT
relationships.related_component_ids	

FA545 — STOP VALVE (GLOBE)

id	FA545
component_name	FA545 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA545
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	80
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA545_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA545_IN, PIPE_FA545_OUT
relationships.related_component_ids	

FA515 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA515
component_name	FA515 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA515
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	46.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA515_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA515_IN, PIPE_FA515_OUT
relationships.related_component_ids	

FA544 — GATE VALVE

id	FA544
component_name	FA544 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA544
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	100
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA544_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA544_IN, PIPE_FA544_OUT
relationships.related_component_ids	

FA514 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA514
component_name	FA514 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA514
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	125
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA514_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA514_IN, PIPE_FA514_OUT
relationships.related_component_ids	

FA574 — BALL VALVE

id	FA574
component_name	FA574 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA574
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	150
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA574_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA574_IN, PIPE_FA574_OUT
relationships.related_component_ids	

FA584 — BALL VALVE

id	FA584
component_name	FA584 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA584
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	50
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	36.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA584_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA584_IN, PIPE_FA584_OUT
relationships.related_component_ids	

FA548 — STOP VALVE (GLOBE)

id	FA548
component_name	FA548 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA548
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	65
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA548_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA548_IN, PIPE_FA548_OUT
relationships.related_component_ids	

FA543 — GATE VALVE

id	FA543
component_name	FA543 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA543
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	80
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA543_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA543_IN, PIPE_FA543_OUT
relationships.related_component_ids	

FA513 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA513
component_name	FA513 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA513
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	53.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA513_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA513_IN, PIPE_FA513_OUT
relationships.related_component_ids	

FA583 — BALL VALVE

id	FA583
component_name	FA583 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA583
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	100
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA583_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA583_IN, PIPE_FA583_OUT
relationships.related_component_ids	

FA573 — BALL VALVE

id	FA573
component_name	FA573 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA573
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	125
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	36.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA573_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA573_IN, PIPE_FA573_OUT
relationships.related_component_ids	

FA547 — STOP VALVE (GLOBE)

id	FA547
component_name	FA547 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA547
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	150
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA547_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA547_IN, PIPE_FA547_OUT
relationships.related_component_ids	

FB544 — STOP VALVE (GLOBE)

id	FB544
component_name	FB544 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB544
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	50
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	13.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB544_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB544_IN, PIPE_FB544_OUT
relationships.related_component_ids	

FB582 — BALL VALVE

id	FB582
component_name	FB582 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB582
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	24.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB582_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB582_IN, PIPE_FB582_OUT
relationships.related_component_ids	

FB572 — BALL VALVE

id	FB572
component_name	FB572 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB572
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	80
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB572_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB572_IN, PIPE_FB572_OUT
relationships.related_component_ids	

FB512 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FB512
component_name	FB512 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB512
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB512_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB512_IN, PIPE_FB512_OUT
relationships.related_component_ids	

FB542 — GATE VALVE

id	FB542
component_name	FB542 - GATE VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB542
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	100
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	20.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB542_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB542_IN, PIPE_FB542_OUT
relationships.related_component_ids	

FB541 — GATE VALVE

id	FB541
component_name	FB541 - GATE VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB541
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	125
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB541_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB541_IN, PIPE_FB541_OUT
relationships.related_component_ids	

FB511 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FB511
component_name	FB511 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB511
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	150
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB511_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB511_IN, PIPE_FB511_OUT
relationships.related_component_ids	

FB571 — BALL VALVE

id	FB571
component_name	FB571 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB571
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	50
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB571_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB571_IN, PIPE_FB571_OUT
relationships.related_component_ids	

FB581 — BALL VALVE

id	FB581
component_name	FB581 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB581
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	36.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB581_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB581_IN, PIPE_FB581_OUT
relationships.related_component_ids	

FB543 — STOP VALVE (GLOBE)

id	FB543
component_name	FB543 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB543
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	80
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB543_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB543_IN, PIPE_FB543_OUT
relationships.related_component_ids	

FA516 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA516
component_name	FA516 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA516
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	25.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA516_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA516_IN, PIPE_FA516_OUT
relationships.related_component_ids	

FA521 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA521
component_name	FA521 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA521
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	32.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA521_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA521_IN, PIPE_FA521_OUT
relationships.related_component_ids	

FA520 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA520
component_name	FA520 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA520
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	125
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA520_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA520_IN, PIPE_FA520_OUT
relationships.related_component_ids	

FA518 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA518
component_name	FA518 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA518
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	150
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	46.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA518_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA518_IN, PIPE_FA518_OUT
relationships.related_component_ids	

FA519 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA519
component_name	FA519 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA519
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	50
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	53.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA519_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA519_IN, PIPE_FA519_OUT
relationships.related_component_ids	

FA517 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA517
component_name	FA517 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA517
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	65
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA575_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA575_IN, PIPE_FA575_OUT
relationships.related_component_ids	

FA575 — SELF-CLOSING VALVE

id	FA575
component_name	FA575- SELF-CLOSING VALVE
component_type	Valve
identification.system_name	Marine Diesel Oil transfer branch (mock assignment; verify service)
identification.location	Fuel Transfer Piping
identification.line_number	FA575
specifications.valve_type	SELF-CLOSING VALVE
specifications.nominal_diameter	65
specifications.material	Body: bronze; stem: stainless steel; seat: PTFE
specifications.connection_type	Flanged end
specifications.operation_type	Spring-return self-closing mechanism
specifications.pressure_rating	10
specifications.valve_opening	100
specifications.flow_rate	15.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Self-Closing Valve for Marine Diesel Oil transfer branch (mock assignment; verify service).
troubleshooting.fault_code	65A_F001
troubleshooting.failure_condition	Return spring failure / incomplete closure
troubleshooting.symptoms	Fails to close, slow return, seat leakage, mechanism sticks open.
troubleshooting.possible_causes	Weak/broken spring, fouled hinge or stem, damaged seat, corrosion, incorrect installation.
troubleshooting.diagnostic_steps	Verify tag and service, inspect return mechanism, test only by approved method, inspect for leakage and obstruction.
troubleshooting.immediate_action	Notify engineer, never wedge or bypass the self-closing action, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep mechanism clean, inspect spring and seat, follow maker instructions and planned maintenance.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_65A_IN, PIPE_65A_OUT
relationships.related_component_ids	

FB112 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FB112
component_name	FB112 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB112
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	32.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB112_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB112_IN, PIPE_FB112_OUT
relationships.related_component_ids	

FB111 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FB111
component_name	FB111 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB111
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB111_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB111_IN, PIPE_FB111_OUT
relationships.related_component_ids	

FA001 — SAFETY / RELIEF VALVE (GLOBE / ANGLE)

id	FA001
component_name	FA001 - SAFETY / RELIEF VALVE (GLOBE / ANGLE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB105
specifications.valve_type	SAFETY / RELIEF VALVE (GLOBE / ANGLE)
specifications.nominal_diameter	65
specifications.material	Body: cast steel; spring: alloy steel; trim: stainless steel
specifications.connection_type	Flanged inlet and discharge
specifications.operation_type	Spring-loaded automatic
specifications.pressure_rating	8
specifications.valve_opening	100
specifications.flow_rate	15.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Safety / Relief Valve (Globe / Angle) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB105_F001
troubleshooting.failure_condition	Failure to lift / failure to reseat
troubleshooting.symptoms	Unwanted discharge, failure to lift at set pressure, failure to reseat, chatter or outlet leakage.
troubleshooting.possible_causes	Incorrect set pressure, contaminated seat, spring deterioration, blocked discharge path, damaged trim.
troubleshooting.diagnostic_steps	Treat discharge as a safety event, verify pressure indication and discharge route, compare against certified set-pressure record. Do not adjust in service.
troubleshooting.immediate_action	Alert responsible engineer immediately, keep clear of discharge, never gag, isolate or adjust except under authorized procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain certification and inspection records, keep discharge path clear, arrange authorized periodic testing.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB105_IN, PIPE_FB105_OUT
relationships.related_component_ids	

FA002— SAFETY / RELIEF VALVE (GLOBE / ANGLE)

id	FA002
component_name	FA002 - SAFETY / RELIEF VALVE (GLOBE / ANGLE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FA002
specifications.valve_type	SAFETY / RELIEF VALVE (GLOBE / ANGLE)
specifications.nominal_diameter	65
specifications.material	Body: cast steel; spring: alloy steel; trim: stainless steel
specifications.connection_type	Flanged inlet and discharge
specifications.operation_type	Spring-loaded automatic
specifications.pressure_rating	8
specifications.valve_opening	100
specifications.flow_rate	20.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Safety / Relief Valve (Globe / Angle) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB104_F001
troubleshooting.failure_condition	Failure to lift / failure to reseat
troubleshooting.symptoms	Unwanted discharge, failure to lift at set pressure, failure to reseat, chatter or outlet leakage.
troubleshooting.possible_causes	Incorrect set pressure, contaminated seat, spring deterioration, blocked discharge path, damaged trim.
troubleshooting.diagnostic_steps	Treat discharge as a safety event, verify pressure indication and discharge route, compare against certified set-pressure record. Do not adjust in service.
troubleshooting.immediate_action	Alert responsible engineer immediately, keep clear of discharge, never gag, isolate or adjust except under authorized procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain certification and inspection records, keep discharge path clear, arrange authorized periodic testing.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB104_IN, PIPE_FB104_OUT
relationships.related_component_ids	

M.D.O S TOR.TK(P) — STORAGE TANK

id	M.D.O S TOR.TK(P)
component_name	M.D.O STORAGE TANK (P)
component_type	Fuel Storage Tank
identification.system_name	Diesel Oil (D.O) Storage System
identification.location	Port Side
identification.line_number	D.O tank connection
specifications.tank_diameter	1800 mm
specifications.material	Marine-grade welded steel
specifications.tank_capacity	5000 L
specifications.tank_pressure	0.01 MPa
specifications.flow_rate	5.0 m3/h
usecase	Storage and transfer of marine diesel oil to the connected D.O system
troubleshooting.fault_code	MDO_TK_P_F001
troubleshooting.failure_condition	Tank not filling, tank overflow, fuel leakage, slow filling, abnormal pressure or vacuum, incorrect level indication
troubleshooting.symptoms	No level increase during transfer; high-level alarm; visible fuel seepage; unusually long filling time; abnormal pressure indication; mismatch between local and remote level readings
troubleshooting.possible_causes	Closed or obstructed transfer path; faulty level sensor or alarm; leaking tank boundary or pipe joint; restricted strainer or reduced pump delivery; restricted vent; faulty level gauge or transmitter
troubleshooting.diagnostic_steps	Confirm valve lineup and transfer source; compare local and remote level readings; inspect tank boundaries and pipe joints; check pump and strainer condition; verify vent and alarm status
troubleshooting.immediate_action	For suspected overflow or leakage, stop transfer and isolate according to the vessel's approved procedure; notify the responsible engineer and follow spill and fire safety procedures
troubleshooting.operational_impact	Reduced D.O storage availability, transfer interruption, potential fuel spill, or downstream supply disruption
troubleshooting.verification_steps	After maintenance, check for leakage, compare level indication with an independent reference, verify alarm operation, and perform a controlled transfer test
troubleshooting.prevention_measures	Inspect tank boundaries, vents, valves, pipe joints, level instruments and alarms routinely; maintain strainers and equipment according to maker instructions
troubleshooting.related_component_ids	FB112, FB001
simulation.current_status	STOPPED
simulation.current_level	0
simulation.commanded_status	STOPPED
simulation.flow_status	NO_FLOW
simulation.flow_rate	0.0
simulation.pressure	0.0
simulation.fault_status	False
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	D.O tank connection pipes
relationships.related_component_ids	FB112, FB001

M.O.D S TOR TK(S) — STORAGE TANK

id	M.O.D S TOR TK(S)
component_name	M.O.D STORAGE TANK (S)
component_type	Fuel Storage Tank
identification.system_name	Diesel Oil (D.O) Storage System
identification.location	Starboard Side
identification.line_number	D.O tank connection
specifications.tank_diameter	1800 mm
specifications.material	Marine-grade welded steel
specifications.tank_capacity	5000 L
specifications.tank_pressure	0.01 MPa
specifications.flow_rate	5.0 m3/h
usecase	Storage and transfer of marine diesel oil to the connected D.O system
troubleshooting.fault_code	MDO_TK_S_F001
troubleshooting.failure_condition	Tank not filling, tank overflow, fuel leakage, slow filling, abnormal pressure or vacuum, incorrect level indication
troubleshooting.symptoms	No level increase during transfer; high-level alarm; visible fuel seepage; unusually long filling time; abnormal pressure indication; mismatch between local and remote level readings
troubleshooting.possible_causes	Closed or obstructed transfer path; faulty level sensor or alarm; leaking tank boundary or pipe joint; restricted strainer or reduced pump delivery; restricted vent; faulty level gauge or transmitter
troubleshooting.diagnostic_steps	Confirm valve lineup and transfer source; compare local and remote level readings; inspect tank boundaries and pipe joints; check pump and strainer condition; verify vent and alarm status
troubleshooting.immediate_action	For suspected overflow or leakage, stop transfer and isolate according to the vessel's approved procedure; notify the responsible engineer and follow spill and fire safety procedures
troubleshooting.operational_impact	Reduced D.O storage availability, transfer interruption, potential fuel spill, or downstream supply disruption
troubleshooting.verification_steps	After maintenance, check for leakage, compare level indication with an independent reference, verify alarm operation, and perform a controlled transfer test
troubleshooting.prevention_measures	Inspect tank boundaries, vents, valves, pipe joints, level instruments and alarms routinely; maintain strainers and equipment according to maker instructions
troubleshooting.related_component_ids	FB111, FB002
simulation.current_status	STOPPED
simulation.current_level	0
simulation.commanded_status	STOPPED
simulation.flow_status	NO_FLOW
simulation.flow_rate	0.0
simulation.pressure	0.0
simulation.fault_status	False
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	D.O tank connection pipes
relationships.related_component_ids	FB111, FB002

D.O S TOR.TK(P) — D.O STORAGE TANK (P)

id	D.O S TOR.TK(P)
component_name	D.O STORAGE TANK (P)
component_type	Fuel Storage Tank
identification.system_name	Diesel Oil (D.O) Storage and Transfer System
identification.location	Port Side
identification.line_number	D.O transfer line; exact line number to be confirmed from vessel piping diagram
specifications.tank_diameter	1800 mm
specifications.material	Marine-grade welded steel
specifications.tank_capacity	5000 L
specifications.tank_pressure	0.01 MPa(g)
specifications.flow_rate	5.0 m3/h
usecase	Bulk diesel storage and transfer to the D.O service/day tank system.
troubleshooting.fault_code	DO_STOR_P_F001
troubleshooting.failure_condition	Storage tank not filling, overfill, fuel leakage, slow transfer, abnormal pressure or vacuum, or incorrect level indication
troubleshooting.symptoms	No level increase during transfer; high/low-level alarm; visible fuel seepage; unusually long filling time; abnormal pressure indication; mismatch between local and remote level readings
troubleshooting.possible_causes	Transfer valve closed or line restricted; transfer pump or strainer issue; level alarm or sensor fault; vent restriction; tank or flange leakage
troubleshooting.diagnostic_steps	Confirm approved transfer source and valve lineup; check level trend; inspect tank boundary and pipe joints; review pump and strainer; verify vent and alarm status

troubleshooting.immediate_action	For suspected overfill or leakage, stop transfer and isolate only under the vessel's approved procedure; notify the responsible engineer
troubleshooting.operational_impact	Loss of bulk D.O reserve and interruption to replenishment of the day/service tank
troubleshooting.verification_steps	After authorized maintenance, check for leakage, compare level indication with an independent reference, verify alarm operation, and perform a controlled transfer test
troubleshooting.prevention_measures	Inspect tank boundaries, vents, valves, pipe joints, level instruments and alarms routinely; maintain transfer equipment according to maker instructions
troubleshooting.related_component_ids	FB112, FB001
simulation.current_status	STOPPED
simulation.current_level	0
simulation.commanded_status	STOPPED
simulation.flow_status	NO_FLOW
simulation.flow_rate	0.0
simulation.pressure	0.0
simulation.fault_status	False
simulation.fault_code	None
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	D.O transfer piping; exact pipe IDs to be confirmed
relationships.related_component_ids	FB112, FB001

D.O S TOR.TK(S) — D.O STORAGE TANK (S)

id	D.O S TOR.TK(S)
component_name	D.O STORAGE TANK (S)
component_type	Fuel Storage Tank
identification.system_name	Diesel Oil (D.O) Storage and Transfer System
identification.location	Starboard Side
identification.line_number	D.O transfer line; exact line number to be confirmed from vessel piping diagram
specifications.tank_diameter	1800 mm
specifications.material	Marine-grade welded steel
specifications.tank_capacity	5000 L
specifications.tank_pressure	0.01 MPa(g)
specifications.flow_rate	5.0 m3/h
usecase	Bulk diesel storage and transfer to the D.O service/day tank system.
troubleshooting.fault_code	DO_STOR_S_F001
troubleshooting.failure_condition	Storage tank not filling, overfill, fuel leakage, slow transfer, abnormal pressure or vacuum, or incorrect level indication
troubleshooting.symptoms	No level increase during transfer; high/low-level alarm; visible fuel seepage; unusually long filling time; abnormal pressure indication; mismatch between local and remote level readings
troubleshooting.possible_causes	Transfer valve closed or line restricted; transfer pump or strainer issue; level alarm or sensor fault; vent restriction; tank or flange leakage
troubleshooting.diagnostic_steps	Confirm approved transfer source and valve lineup; check level trend; inspect tank boundary and pipe joints; review pump and strainer; verify vent and alarm status
troubleshooting.immediate_action	For suspected overfill or leakage, stop transfer and isolate only under the vessel's approved procedure; notify the responsible engineer
troubleshooting.operational_impact	Loss of bulk D.O reserve and interruption to replenishment of the day/service tank
troubleshooting.verification_steps	After authorized maintenance, check for leakage, compare level indication with an independent reference, verify alarm operation, and perform a controlled transfer test
troubleshooting.prevention_measures	Inspect tank boundaries, vents, valves, pipe joints, level instruments and alarms routinely; maintain transfer equipment according to maker instructions
troubleshooting.related_component_ids	FB111, FB002
simulation.current_status	STOPPED
simulation.current_level	0
simulation.commanded_status	STOPPED
simulation.flow_status	NO_FLOW
simulation.flow_rate	0.0
simulation.pressure	0.0
simulation.fault_status	False
simulation.fault_code	None
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	D.O transfer piping; exact pipe IDs to be confirmed
relationships.related_component_ids	FB111, FB002

D.O S DAY.TK(P) — D.O DAY TANK (P)

id	D.O S DAY.TK(P)
component_name	D.O DAY TANK (P)
component_type	Fuel Day Tank
identification.system_name	Diesel Oil (D.O) Storage and Transfer System
identification.location	Port Side
identification.line_number	D.O transfer line; exact line number to be confirmed from vessel piping diagram
specifications.tank_diameter	1000 mm
specifications.material	Marine-grade welded steel
specifications.tank_capacity	1000 L
specifications.tank_pressure	0.01 MPa(g)
specifications.flow_rate	2.0 m3/h
usecase	Daily-use diesel buffer tank supplying connected D.O consumers on the Port side, subject to vessel arrangement.
troubleshooting.fault_code	DO_DAY_P_F001
troubleshooting.failure_condition	Day tank not replenishing, overfill, low-level condition, leakage, unstable level, or supply interruption
troubleshooting.symptoms	No level increase during transfer; high/low-level alarm; visible fuel seepage; unusually long filling time; abnormal pressure indication; mismatch between local and remote level readings
troubleshooting.possible_causes	Replenishment valve or pump unavailable; level control malfunction; transfer line restriction; high/low level switch fault; vent restriction; leakage
troubleshooting.diagnostic_steps	Check day-tank level and alarms; verify approved replenishment sequence and valve lineup; confirm transfer pump availability; inspect line and vent; compare local and remote level indications
troubleshooting.immediate_action	If low level threatens consumers, notify the responsible engineer and follow the approved replenishment procedure; stop transfer for overfill or leakage
troubleshooting.operational_impact	Reduced D.O buffer time and possible interruption to connected machinery or consumers
troubleshooting.verification_steps	After authorized maintenance, check for leakage, compare level indication with an independent reference, verify alarm operation, and perform a controlled transfer test
troubleshooting.prevention_measures	Inspect tank boundaries, vents, valves, pipe joints, level instruments and alarms routinely; maintain transfer equipment according to maker instructions
troubleshooting.related_component_ids	Not assigned; confirm from approved piping diagram
simulation.current_status	STOPPED
simulation.current_level	0
simulation.commanded_status	STOPPED
simulation.flow_status	NO_FLOW
simulation.flow_rate	0.0
simulation.pressure	0.0
simulation.fault_status	False
simulation.fault_code	None
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	D.O transfer piping; exact pipe IDs to be confirmed
relationships.related_component_ids	

D.O S DAY.TK(S) — D.O DAY TANK (S)

id	D.O S DAY.TK(S)
component_name	D.O DAY TANK (S)
component_type	Fuel Day Tank
identification.system_name	Diesel Oil (D.O) Storage and Transfer System
identification.location	Starboard Side
identification.line_number	D.O transfer line; exact line number to be confirmed from vessel piping diagram
specifications.tank_diameter	1000 mm
specifications.material	Marine-grade welded steel
specifications.tank_capacity	1000 L
specifications.tank_pressure	0.01 MPa(g)
specifications.flow_rate	2.0 m3/h
usecase	Daily-use diesel buffer tank supplying connected D.O consumers on the Starboard side, subject to vessel arrangement.
troubleshooting.fault_code	DO_DAY_S_F001
troubleshooting.failure_condition	Day tank not replenishing, overfill, low-level condition, leakage, unstable level, or supply interruption
troubleshooting.symptoms	No level increase during transfer; high/low-level alarm; visible fuel seepage; unusually long filling time; abnormal pressure indication; mismatch between local and remote level readings
troubleshooting.possible_causes	Replenishment valve or pump unavailable; level control malfunction; transfer line restriction; high/low level switch fault; vent restriction; leakage
troubleshooting.diagnostic_steps	Check day-tank level and alarms; verify approved replenishment sequence and valve lineup; confirm transfer pump availability; inspect line and vent; compare local and remote level indications
troubleshooting.immediate_action	If low level threatens consumers, notify the responsible engineer and follow the approved replenishment procedure; stop transfer for overfill or leakage
troubleshooting.operational_impact	Reduced D.O buffer time and possible interruption to connected machinery or consumers
troubleshooting.verification_steps	After authorized maintenance, check for leakage, compare level indication with an independent reference, verify alarm operation, and perform a controlled transfer test
troubleshooting.prevention_measures	Inspect tank boundaries, vents, valves, pipe joints, level instruments and alarms routinely; maintain transfer equipment according to maker instructions
troubleshooting.related_component_ids	Not assigned; confirm from approved piping diagram
simulation.current_status	STOPPED
simulation.current_level	0
simulation.commanded_status	STOPPED
simulation.flow_status	NO_FLOW
simulation.flow_rate	0.0
simulation.pressure	0.0
simulation.fault_status	False
simulation.fault_code	None
simulation.animation_state	IDLE
simulation.operating_hours	0
relationships.connected_pipe_ids	D.O transfer piping; exact pipe IDs to be confirmed
relationships.related_component_ids	